Exploring the Best Frameworks FOR RESPONSIVE WEB DESIGN

Responsive Web design is what makes pages render well on a variety of devices and screen sizes. In the current scenario, every customer expects a responsive website that is compatible with all devices, especially mobiles and smartphones. This article takes a quick look at some of the most popular RWD frameworks.

Responsive Web Design (RWD) involves the use of HTML and CSS to automatically resize, hide, shrink and enlarge a website to make it look good on all devices—desktops, tablets and mobiles.

A responsive design uses a fluid grid structure so that all page elements are sized by proportion rather than pixels. So if you have a four-column structure in your page, we can divide the website into a four-grid structure (how the website is rendered also depends on every column's width). Suppose we have two columns that are wider than the other two, we divide the first two columns across 40 per cent of the width of the screen and the other two across 20 per cent of the screen's width.

In responsive design, the image is sized for clarity on all devices. The viewport is the most determining factor of RWD. A viewport is the visible area of a Web page on the user's device. In creating a RWD, the header of the HTML page mentions the viewport in the *meta* tag as:

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<meta name="viewport" content="width=device-width, initial-scale=1.0">
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A <*meta*> *viewport* element gives the browser instructions on how to control the page's dimensions and scaling.

The 'width=device-width' value sets the width of the page to adjust the screen-width of the device (which will differ depending on the device type). The 'initial-scale=1.0' value sets the initial zoom level when the page is first loaded by the browser.

In responsive design, the grid layout, which often has 12 columns and has a total width of 100 per cent, is very helpful; it automatically shrinks and resizes according to the browser window width. For design, any RWD gives first preference to mobiles, then to tablets and after that, to desktops.